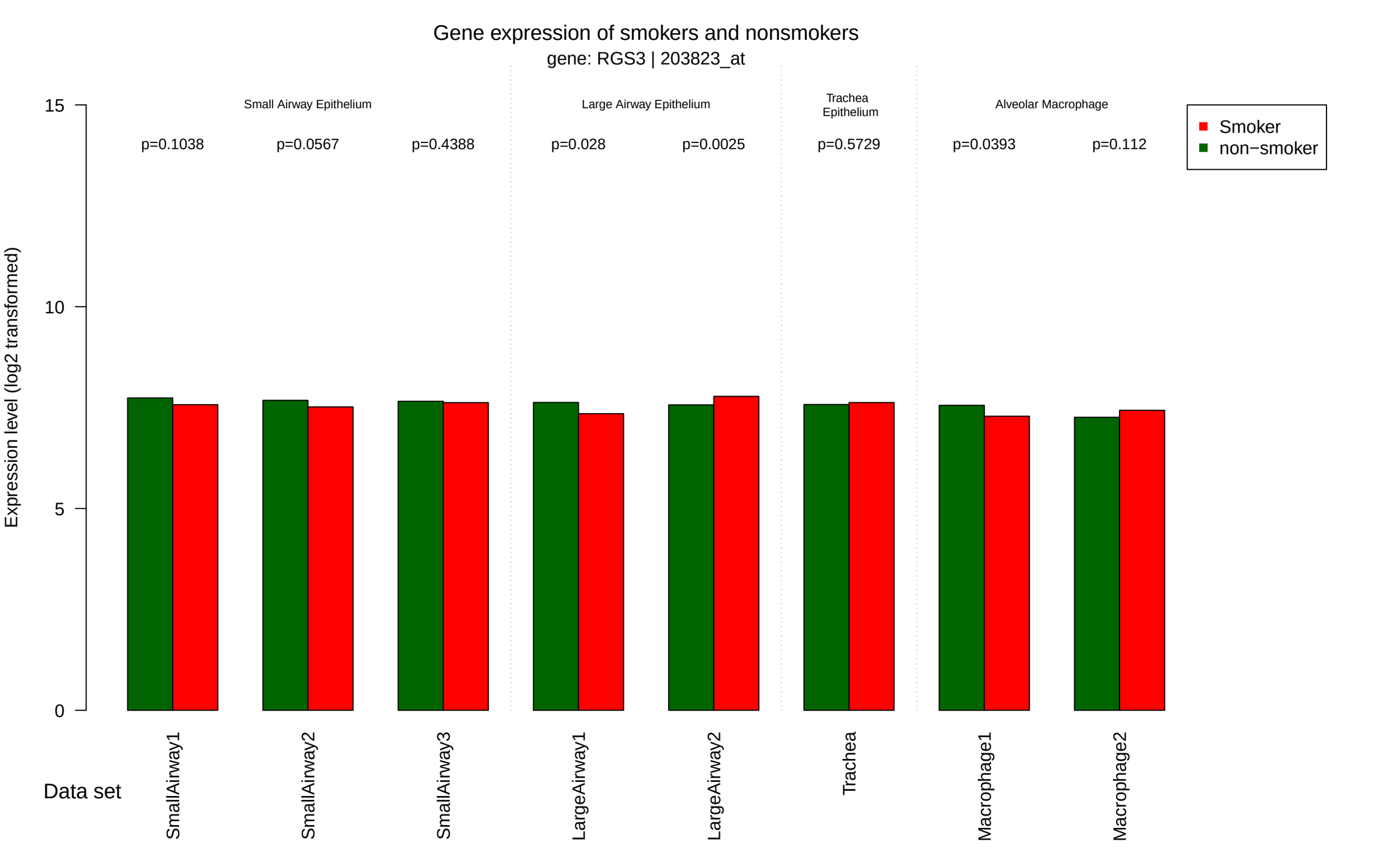
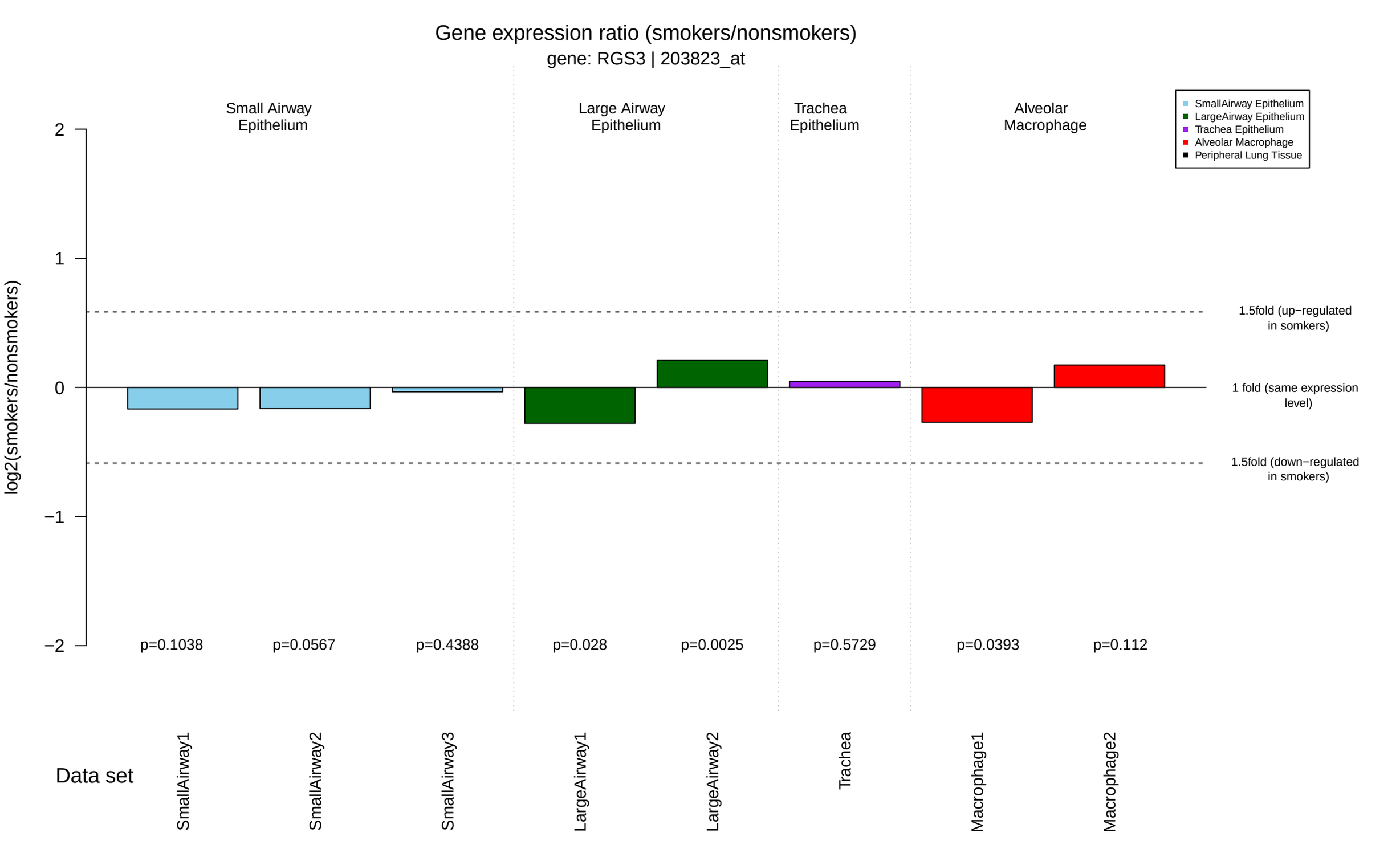
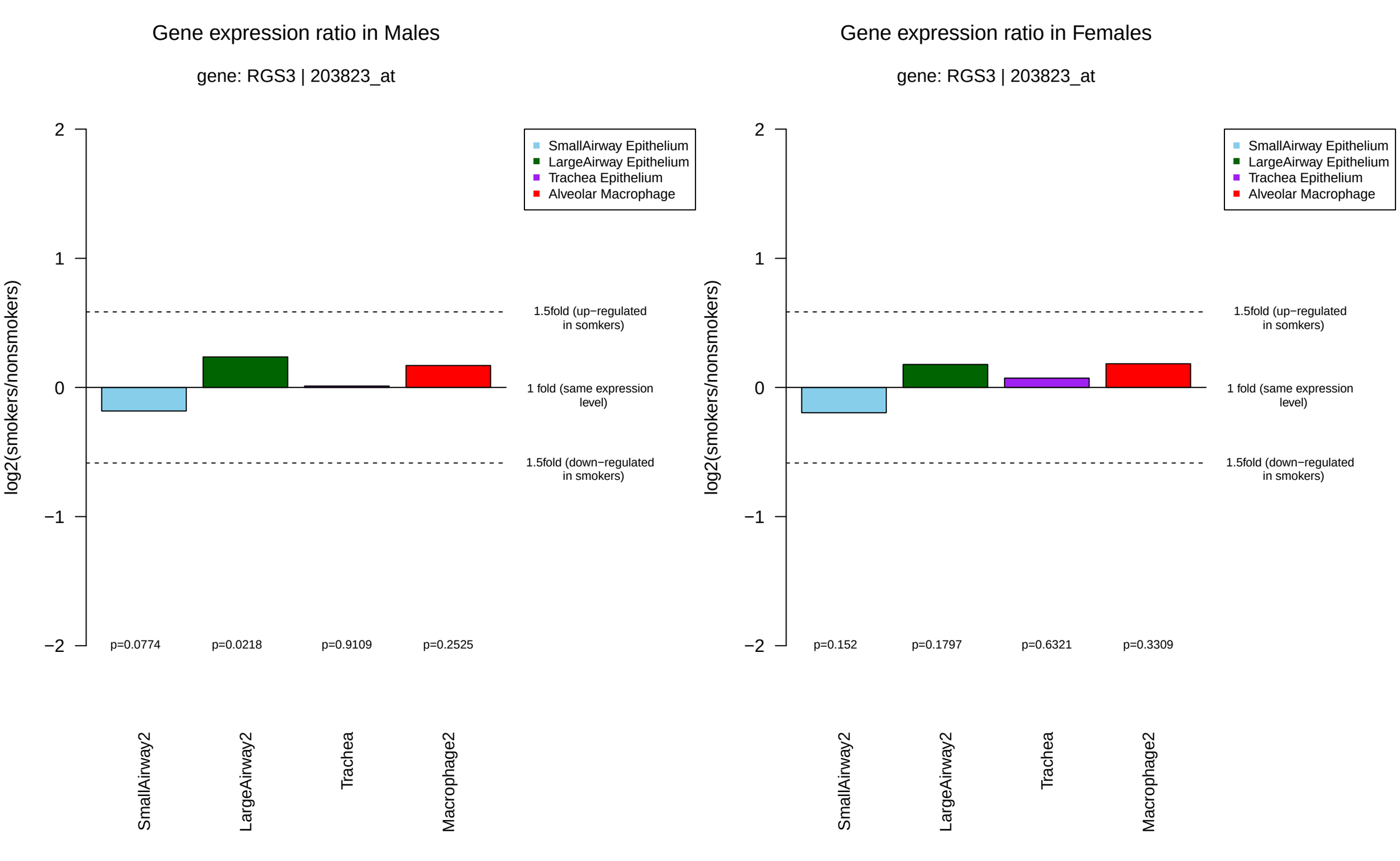
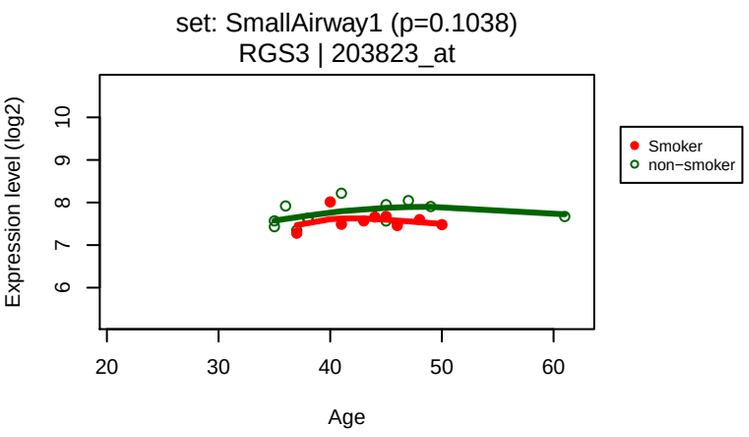
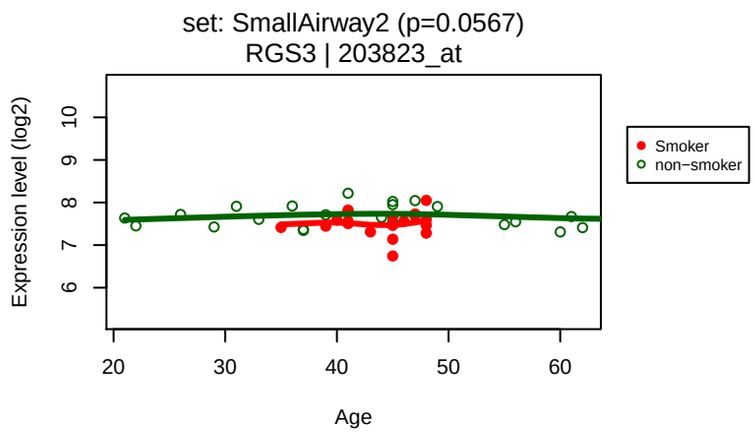
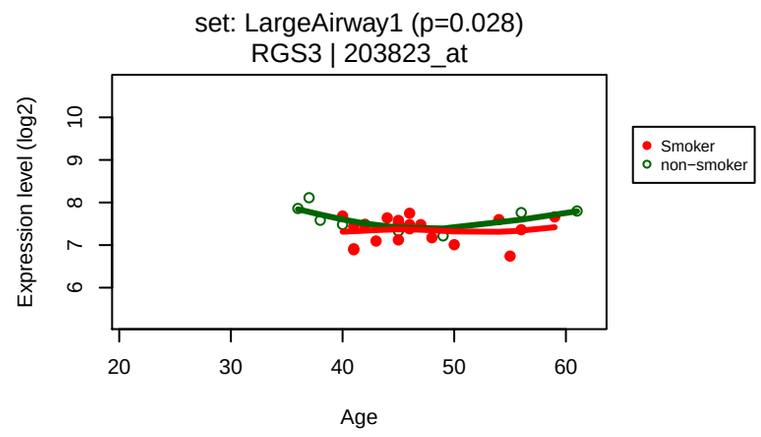

Gene expression of smokers and nonsmokers
gene: RGS3 | 203823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1038
p=0.0567
p=0.4388
p=0.028
p=0.0025
p=0.5729
p=0.0393
p=0.112
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

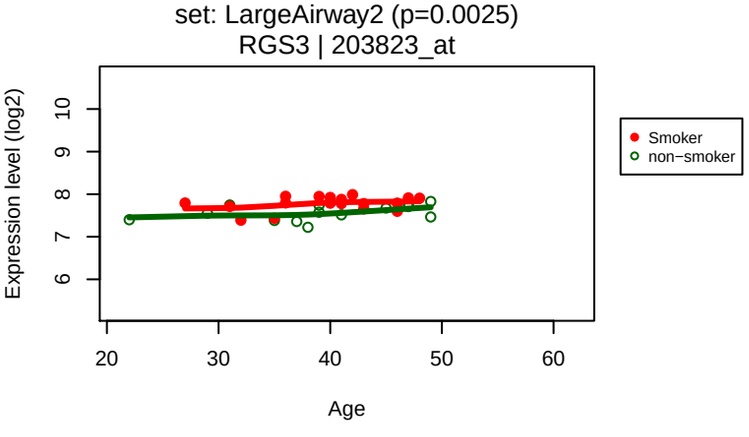

Gene expression ratio (smokers/nonsmokers)
gene: RGS3 | 203823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1038
p=0.0567
p=0.4388
p=0.028
p=0.0025
p=0.5729
p=0.0393
p=0.112
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

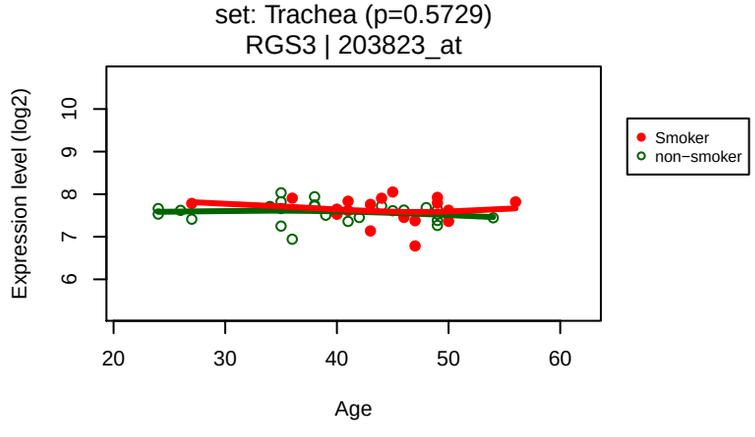

Gene expression ratio in Males
gene: RGS3 | 203823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0774
p=0.0218
p=0.9109
p=0.2525
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS3 | 203823_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.152
p=0.1797
p=0.6321
p=0.3309
SmallAirway2
LargeAirway2
Trachea
Macrophage2

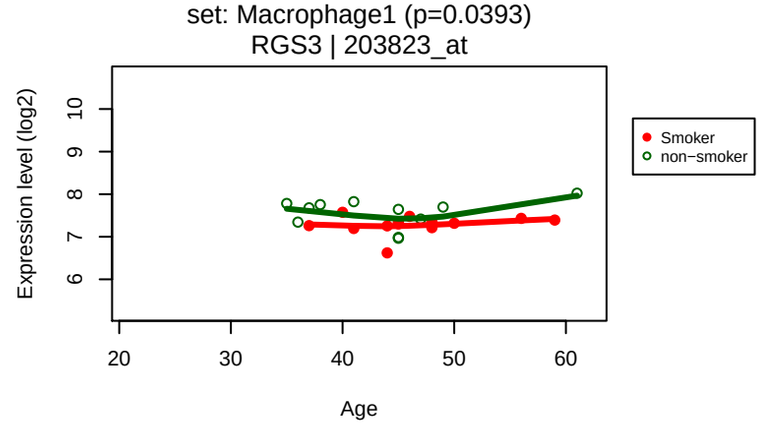

set: SmallAirway1 (p=0.1038)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker

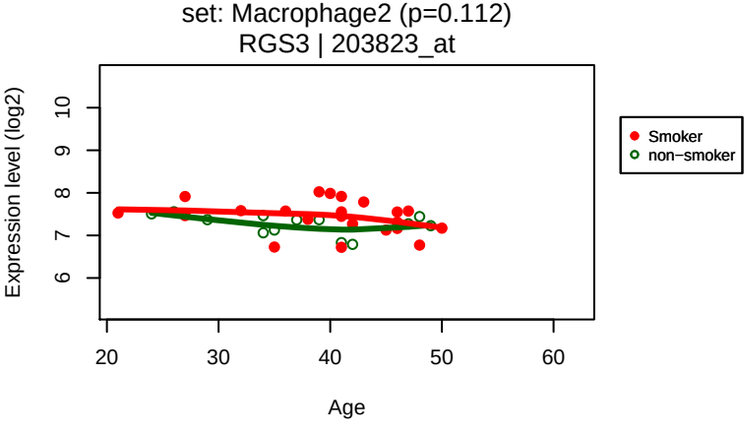

set: SmallAirway2 (p=0.0567)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker

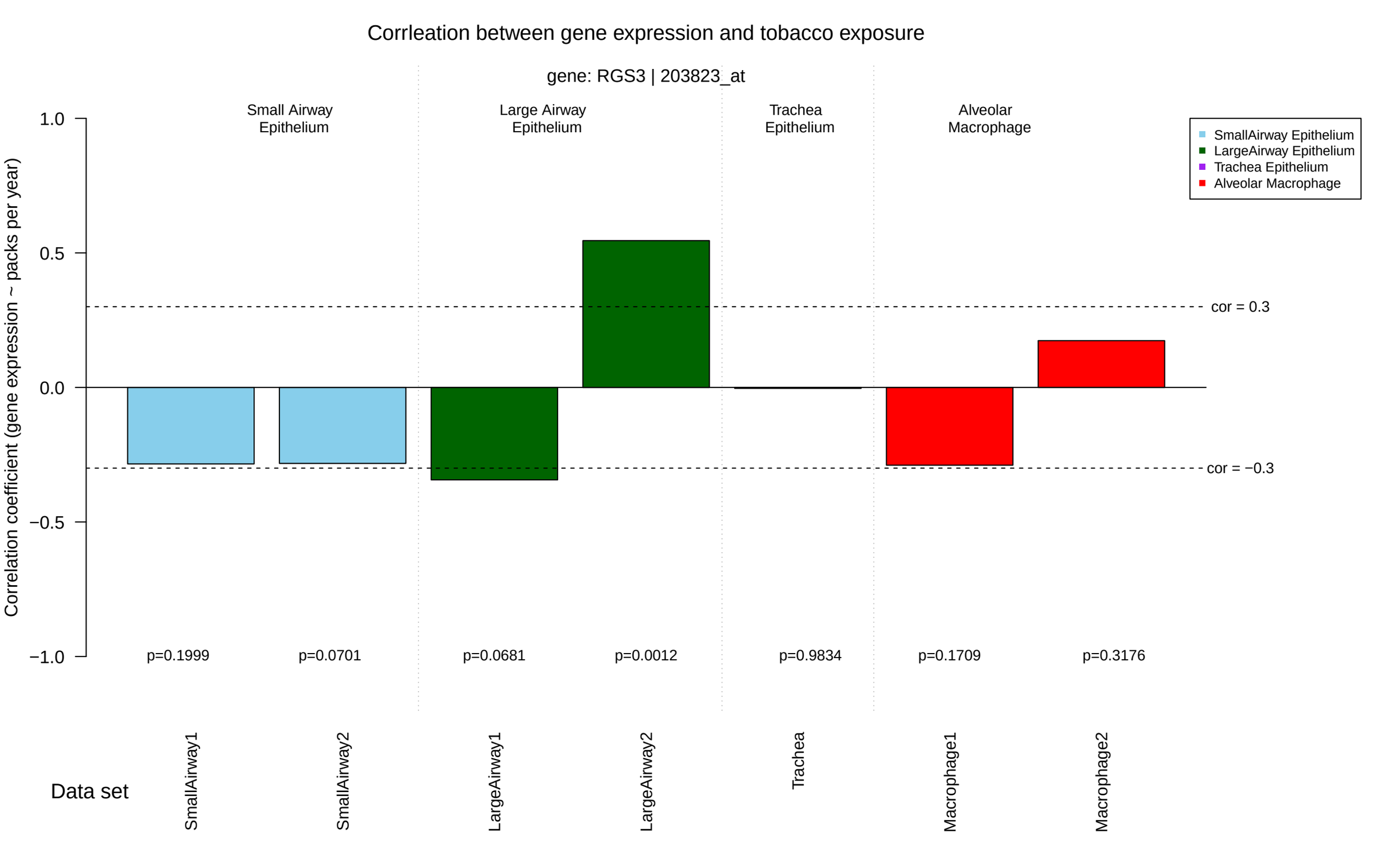

set: LargeAirway1 (p=0.028)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0025)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5729)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0393)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.112)
RGS3 | 203823_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS3 | 203823_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1999
p=0.0701
p=0.0681
p=0.0012
p=0.9834
p=0.1709
p=0.3176
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2